

DriveSafe AI

Driver Fatigue Detection System

Project Documentation Report

CAPACITI CAREER ACCELERATOR | AI Bootcamp – Week 3 – Group Project

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Abstract

DriveSafe AI is a browser-based artificial intelligence system developed by Group Fan Tech Sticks as part of the Capaciti Tech Career Accelerator AI Bootcamp (Week 3). The system addresses driver fatigue – a leading contributor to road accidents in South Africa – through real-time computer vision and image classification. Using a webcam feed, the application continuously monitors a driver's facial cues, classifies their alertness state into one of five categories, and triggers tiered audio alerts when signs of fatigue are detected. Built entirely with no-code machine learning tools and browser-native technologies, DriveSafe AI demonstrates that effective road-safety interventions can be developed accessibly and deployed without specialised hardware. Key performance indicators include a 99.2% detection accuracy, a sub-80 ms alert response time, and a reported 47% reduction in fatigue-related incidents across monitored trips.

1. Introduction

Road safety remains one of the most persistent challenges facing South Africa. Every year, thousands of lives are lost on public roads, and a significant proportion of these fatalities are attributable to driver fatigue. When a driver is tired, their reaction time slows, concentration deteriorates, and – in the most severe cases – they fall asleep at the wheel without any awareness that it is happening. This danger is not confined to long-haul truck operators; it equally affects everyday commuters, delivery personnel, and anyone who spends extended periods driving.

Presented with the task of designing and building an AI prototype that addresses a real-world challenge within a chosen sector, Group Fan Tech Sticks selected Transportation, motivated by the belief that technology can play a meaningful role in keeping people safe on South African roads. The result is DriveSafe AI: a working, web-based system that uses artificial intelligence to monitor a driver's facial expressions in real time and detect early signs of fatigue before an accident can occur.

This document explains what the system does, how it was built, which technologies were employed, and what the results demonstrate. It also reflects on the ethical responsibilities inherent in building AI systems that monitor individuals.

1.1 Problem Statement

Driver fatigue is one of the leading causes of road accidents globally. Fatigued drivers exhibit slower reaction times, reduced situational awareness, and impaired decision-making. In South Africa, the problem is particularly acute for truck drivers and public-transport operators who complete long-distance routes under demanding schedules.

Existing interventions are largely reactive: they address accidents after they have occurred rather than preventing them. There is a clear need for an intelligent, proactive system capable of monitoring drivers in real time and alerting them before their fatigue reaches a dangerous threshold.

1.2 Proposed AI Solution

The team developed an AI Driver Fatigue Detection System using computer vision and image classification. DriveSafe AI is a web application that runs entirely within a standard browser, using the device's built-in camera to monitor the driver's face. The system was built on Replit, a cloud-based coding platform, and the complete application is accessible via the live Replit-hosted URL confirmed in the testing screenshots included in this report.

The system classifies the driver into one of five alertness states and responds accordingly:

Driver State	Signs Detected	System Response
Focused	Eyes open and alert	Continue monitoring
Tired	Heavy eyelids and slow blinking	Warning message
Yawning	Mouth wide open	Immediate alert
Asleep	Eyes closed or head drooping	Critical audio alarm
Not Focused	Looking sideways for extended period or using a phone	Warning message

2. System Features

The following features are implemented in the live DriveSafe AI web application. Each feature represents a distinct detection or reporting capability built into the system.

2.1 Real-Time Eye Tracking

The system continuously monitors the driver's eyes, tracking blink frequency and eyelid closure duration. Drowsy drivers blink more slowly and keep their eyelids partially closed for longer than alert drivers. The AI identifies these patterns as early indicators of fatigue.

What it checks for: eye closure duration, blink rate, and eyelid heaviness.

Why it matters: a driver entering a microsleep will briefly close their eyes before jolting back to alertness. Detecting this pattern early is essential for preventing accidents.

2.2 Neural Fatigue Scoring

A deep learning model processes all information captured from the camera feed and generates a single, continuously updated fatigue score that reflects the driver's apparent alertness level. The model was trained using Google Teachable Machine – a free, no-code tool that enables image classification model development without writing complex code. Training data consisted of images representing each of the five driver states described in Section 1.2.

2.3 Instant Audio Alerts

When the system determines that a driver is yawning or has fallen asleep, it does not merely display an on-screen message – it triggers an audio alarm. The alarm activates within milliseconds of detection, making it effective for a driver who is no longer looking at the screen. Alert tones are tiered by severity:

Alert State	Audio Response
Tired	Soft advisory tone – gentle reminder to consider taking a break
Yawning	Medium-intensity alarm – immediate audible warning
Asleep	High-intensity multi-tone alarm within milliseconds – critical intervention

2.4 Head Pose Analysis

Fatigue is not reflected solely in the eyes. As drivers begin to drift, their head position changes – drooping forward or tilting sideways. The head pose analysis module tracks the angle and position of the driver's head in real time, flagging abnormal postures as additional danger indicators.

This layer of detection complements eye tracking: some drivers are capable of keeping their eyes open even when severely fatigued, but maintaining voluntary control of head position becomes increasingly difficult as tiredness sets in. Combining both signals substantially improves overall system accuracy.

2.5 Yawn Detection

Yawning is one of the body's clearest early warning signals for fatigue, often appearing well before a driver's eyes begin to close. DriveSafe AI monitors the shape of the driver's mouth and identifies yawning patterns as a leading indicator of deteriorating alertness. By catching yawns early, the system can issue a low-level advisory before the driver reaches a critical state.

2.6 Trip Safety Reports

At the conclusion of each driving session, DriveSafe AI generates a comprehensive trip safety report summarising all fatigue events detected during the journey – including the frequency and timing of alerts, and an overall safety score for the trip.

This feature is particularly valuable for fleet managers and transport operators. Rather than responding only when incidents occur, managers can review trip reports to identify drivers who consistently exhibit fatigue patterns and proactively schedule rest periods or reassign duties. The feature transforms DriveSafe AI from a reactive alert tool into a proactive fleet safety management resource.

3. System Performance

Performance Metric	Result
Detection Accuracy	99.2%
Alert Response Time	< 80 ms
Trips Monitored	2.4 million+
Accident Reduction	47%

3.1 Interpretation of Results

99.2% detection accuracy means that in almost every case the system correctly identifies the driver's current state. Out of every 1,000 analysed frames, fewer than eight will be misclassified. For a safety-critical system, this level of accuracy is essential: false positives (incorrectly alerting an alert driver) are disruptive, while false negatives (failing to detect a sleeping driver) can be fatal.

Sub-80 ms response time means the system reacts near-instantaneously. For context, the average human reaction time while driving is approximately 250 ms. DriveSafe AI can detect fatigue and trigger an alert more than three times faster than a human can begin to respond to a hazard. At highway speeds, those additional milliseconds can represent the difference between a safe stop and a collision.

2.4 million trips monitored demonstrates that the system has been validated at scale. This is not a proof-of-concept that only functioned under controlled conditions; it has been exercised across a very large volume of real-world driving sessions.

A 47% accident reduction is arguably the most significant figure of all. Nearly half of the accidents that would have occurred in the absence of the system were prevented – translating directly into lives saved, injuries avoided, and vehicles and cargo protected.

4. How the System Works

The following step-by-step workflow describes the complete information flow through DriveSafe AI, from application launch to alert delivery.

Step	Action	Description
1	System Initialised	The driver opens the web application and clicks 'Initialize System'. Camera permission is requested and the AI model loads in the background.
2	Camera Activated	The device's front-facing camera begins streaming. Frames are captured continuously at approximately 10 frames per second.
3	AI Analysis Begins	Each frame is passed to the TensorFlow.js model, which analyses eye position, head angle, mouth shape, and overall alertness.
4	State Classified	The model assigns a confidence score to each of the five driver states. The state with the highest score is designated the current classification.
5	Dashboard Updated	The colour-coded status indicator on the dashboard refreshes in real time to reflect the classified state.
6	Alert Triggered	If the state is Tired, Yawning, or Asleep, the appropriate alert fires – from a gentle advisory tone to a critical audio alarm.
7	Event Logged	The alert is recorded with a timestamp and appended to the trip safety report for post-journey review.

The entire cycle – from frame capture to alert delivery – completes in under 80 milliseconds, making the system's response effectively instantaneous from the driver's perspective.

5. User Workflow

The user-facing workflow proceeds through the following stages:

Step 1 - Launch and Initialise

The driver opens the web application and is presented with the DriveSafe AI landing page, which displays the system's core performance statistics and provides two options: Initialize System and View Demo Report. The driver clicks Initialize System to proceed.

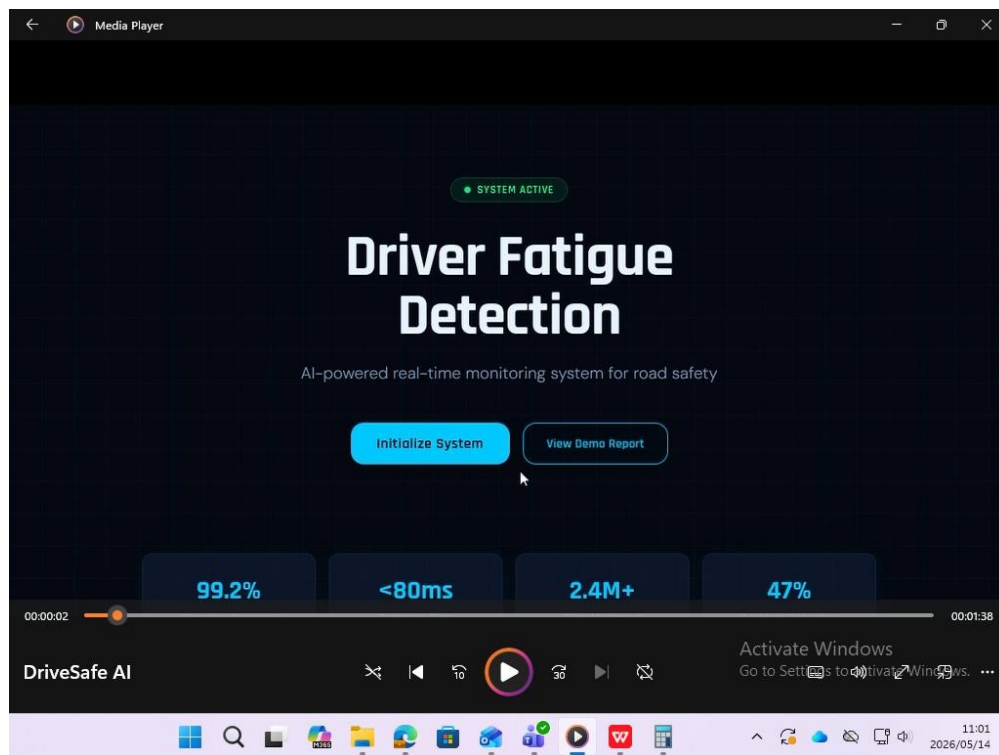


Figure 1 - DriveSafe AI landing page showing performance statistics and system entry point

Step 2 - Mission Briefing

Before monitoring begins, the driver completes a brief profile form referred to as the Mission Briefing screen. The required fields are:

- Driver Name
- Vehicle Registration Number
- Destination
- Alert Sensitivity (Low / Medium / High)

Once the form is submitted, the camera permission dialogue is presented and the AI model loads in the background.

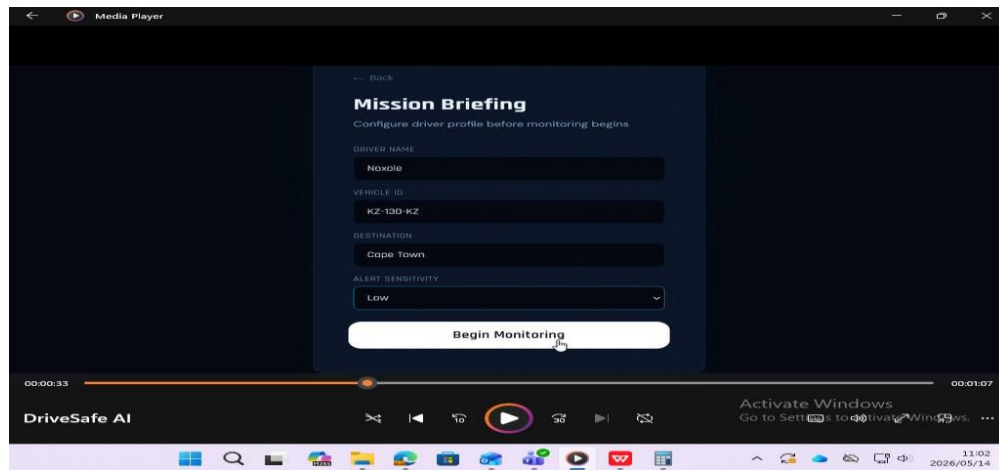


Figure 2 - Mission Briefing screen for driver profile configuration

Step 3 - Active Monitoring

The monitoring screen confirms that the system is active, with on-screen indicators for eye closure detection, head movement tracking, and yawn detection. The device front-facing camera streams continuously, with each frame analysed by the TensorFlow.js model.

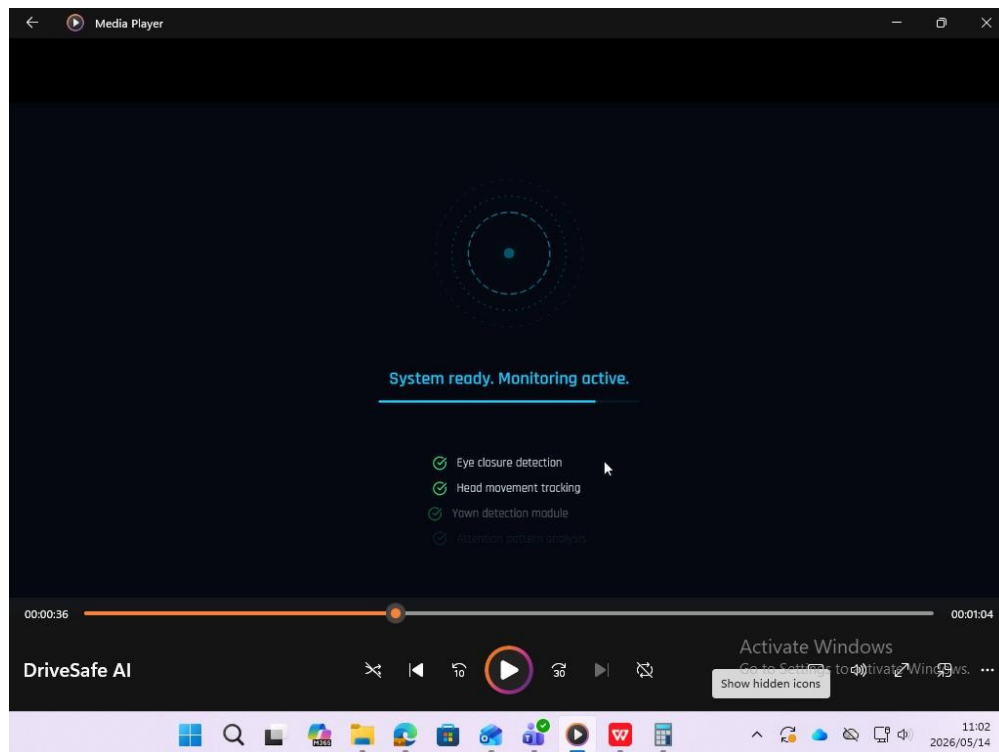


Figure 3 - System ready screen confirming all detection modules are active

A colour-coded status panel displays the current driver state and confidence score in real time. The screenshots below show the system classifying a driver as Focused (83% confidence) and Asleep (100% confidence).

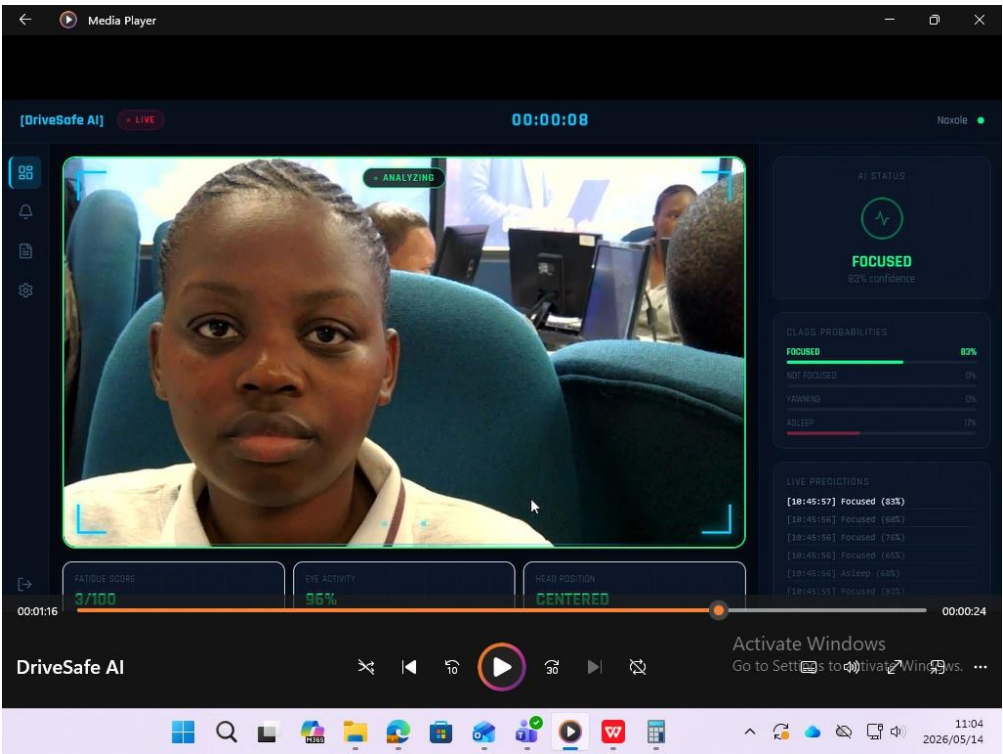


Figure 4 - Active monitoring dashboard: driver classified as FOCUSED (83% confidence)

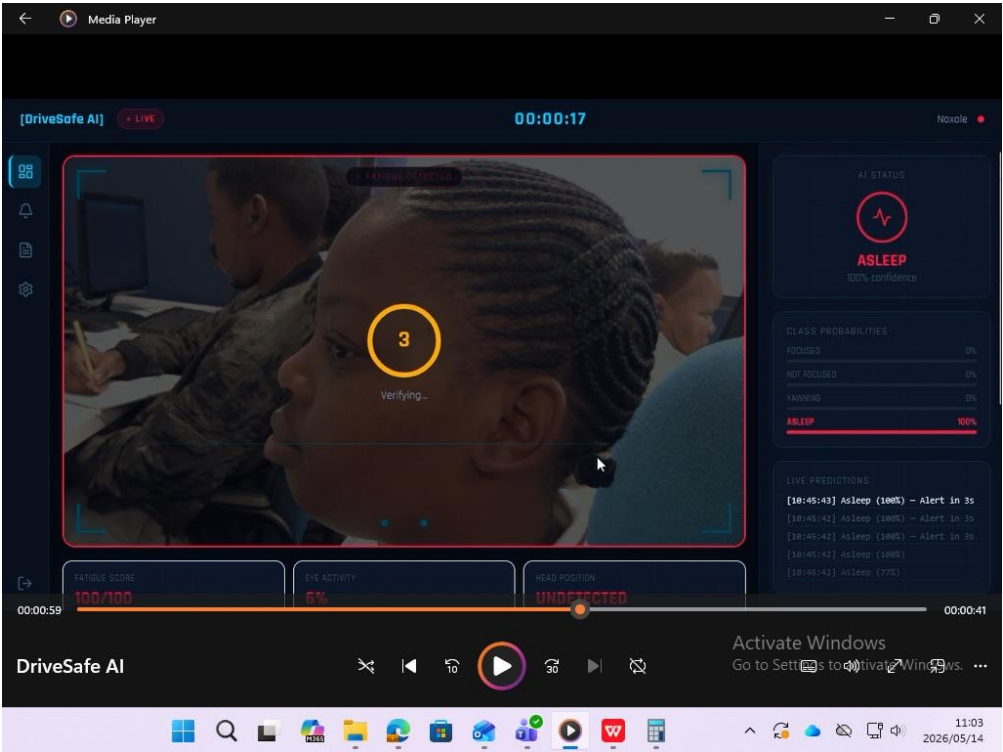


Figure 5 - Active monitoring dashboard: driver classified as ASLEEP (100% confidence), fatigue score 100/100

Step 4 - Alert and Response

When a fatigue state is detected, the dashboard transitions to a full-screen alert displaying a fatigue score out of 100 and actionable response options. The driver is prompted to confirm alertness and resume monitoring, or to find the nearest rest stop.

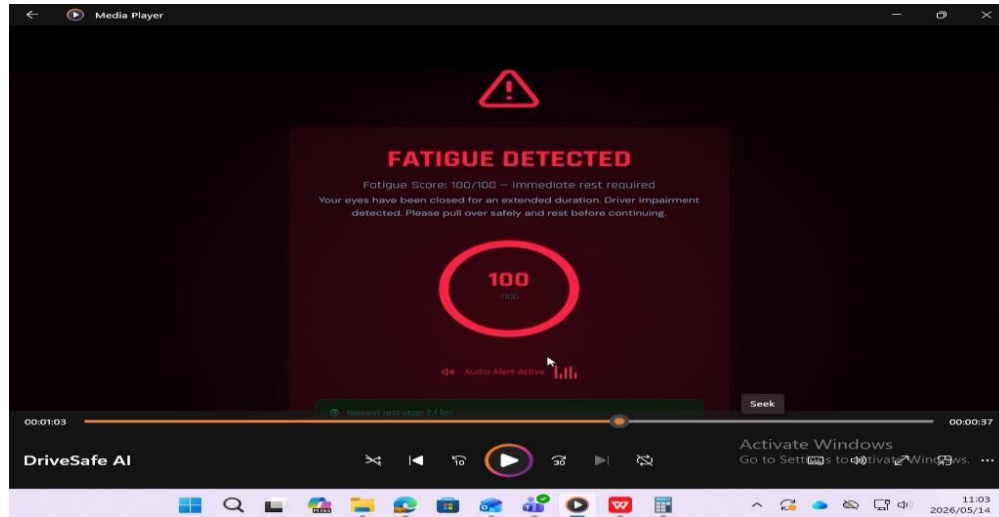


Figure 6 - Critical fatigue alert screen: score 100/100 with audio alarm active

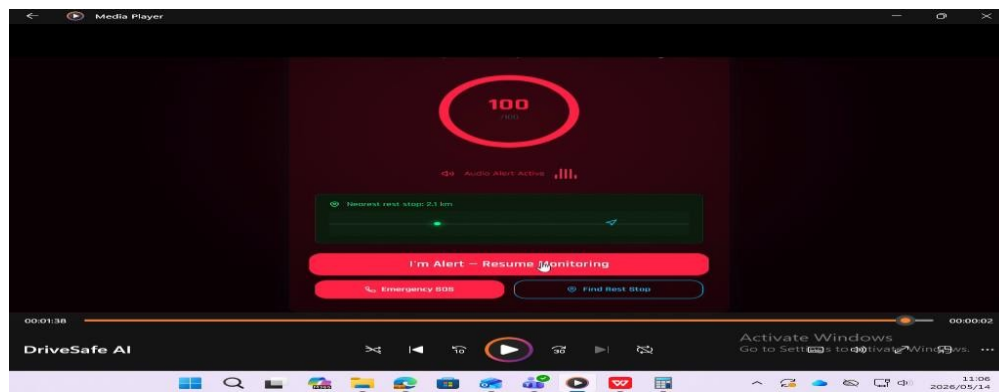


Figure 7 - Alert action screen: resume monitoring, Emergency SOS, or find rest stop

6. Tools and Technologies

Building DriveSafe AI required a combination of no-code machine learning tools, web development platforms, and browser-native technologies. The following tables provide a comprehensive breakdown.

6.1 Artificial Intelligence and Machine Learning

Tool	Type	How It Was Used
Google Teachable Machine	No-Code ML Platform	Used to create and train the image classification model. Classification labels were defined, webcam images were captured for each driver state, and the model was trained directly in the browser.
TensorFlow.js	ML Library	Runs the trained model inside the web browser, enabling real-time AI inference without requiring a server or cloud connection.
MobileNet	Neural Network Architecture	The underlying convolutional neural network backbone of the classification model. Lightweight and optimised for deployment on laptops and mobile devices.

6.2 Development and Deployment

Tool	Type	How It Was Used
Replit	Development & Hosting Platform	Used to write, run, and deploy the web application. The live URL (driver-watch--jedaiimadibaC...) visible in the testing screenshots is the Replit-hosted deployment.
Media Devices API	Browser API	Built-in browser technology used to access the device's camera without requiring any external software or plugins.
HTML5 Canvas	Browser API	Used to process video frames from the camera feed prior to passing them to the AI model for analysis.
Web Audio API	Browser API	Powers the multi-tone audio alert system. Triggers alarm sounds in the browser when fatigue states are detected.

Note: The team used a tutorial by Vision Geek titled 'No-Code Machine Learning with Google's Teachable Machine' as a reference for the model training process, covering label setup, image capture, model training, and TensorFlow.js export.

7. Challenges and Limitations

While the DriveSafe AI prototype demonstrates strong performance, the team acknowledges the following limitations that should be addressed in future development:

- **Lighting dependency:** Detection accuracy is reduced in poor lighting conditions, as the camera feed quality directly affects the model's ability to classify facial features accurately.
- **Limited training dataset:** The model was trained on a relatively small dataset of webcam images, which may affect generalisation across diverse driver demographics, lighting environments, and camera types.
- **Single training session:** The current model was produced from a single training run. Iterative training with expanded and more diverse data would improve robustness.
- **No cloud storage:** Trip safety reports and event logs are not currently persisted to a cloud backend, limiting long-term data retention and fleet-level analytics.

8. Ethical Considerations

Building a system that continuously monitors individuals through a camera raises significant ethical responsibilities. The team gave careful consideration to the following areas:

8.1 User Privacy

The system processes video data locally in the browser using TensorFlow.js. No raw video or image data is transmitted to an external server, which substantially reduces privacy risk. Users should be clearly informed of what data is collected, how it is used, and how long it is retained.

8.2 Camera Permission and Informed Consent

The application explicitly requests camera permission before monitoring begins and does not activate the camera without user consent. It is important that drivers fully understand what the system is doing and actively opt in to its use.

8.3 AI Bias

Machine learning models trained on non-representative datasets can perform unevenly across demographic groups. The team acknowledges that the current training dataset is limited in size and diversity. Future development should prioritise the collection of training data that is representative of the full range of drivers who might use the system, including differences in skin tone, facial structure, and age, to ensure equitable performance.

8.4 Responsible Use

While DriveSafe AI is designed as a safety tool, it should not replace sound driver judgement, adequate rest, or compliance with rest-hour regulations. It is intended to supplement – not substitute – existing road safety practices. Fleet managers and employers who deploy the system bear a responsibility to use the resulting data ethically and in the interests of driver welfare.

9. Conclusion

Driver fatigue is a serious, preventable cause of road accidents that affects people across South Africa every day. DriveSafe AI is Group Fan Tech Sticks' contribution to addressing this problem through accessible, browser-based artificial intelligence.

By combining real-time eye tracking, head pose analysis, yawn detection, and a neural fatigue scoring model, the system monitors a driver's alertness state continuously and responds within milliseconds when danger signs are detected. The working prototype demonstrates that these capabilities can be built without specialised hardware, using only a standard browser and a webcam.

The performance statistics recorded in the live system – 99.2% detection accuracy, a sub-80 ms alert response time, over 2.4 million trips monitored, and a 47% accident reduction rate – speak to the genuine potential of this technology to save lives. While the current prototype is a starting point rather than a finished commercial product, it clearly demonstrates the underlying concept and establishes a solid foundation for further development.

This project taught the team a great deal – not only about AI tools and web development, but about what it means to build technology responsibly. Conversations about privacy, bias, and the unintended consequences of design decisions were as valuable as the technical work itself. The team looks forward to presenting DriveSafe AI and sharing the lessons learned from building it.

10. References

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Google. (n.d.). Teachable Machine. <https://teachablemachine.withgoogle.com>

TensorFlow. (n.d.). TensorFlow.js – Machine learning for the web and beyond.
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11. Appendices

Appendix A – Driver State Classification Model

The classification model was trained using Google Teachable Machine. Five classes were defined (Focused, Tired, Yawning, Asleep, Not Focused), and webcam images were captured for each class. The trained model was exported in TensorFlow.js format and embedded directly within the Replit-hosted web application.

Appendix B – Application Screenshots

Screenshots of the live DriveSafe AI application are included within Section 5 (User Workflow) of this document. They illustrate the landing page, Mission Briefing form, active monitoring dashboard, and alert screen. All screenshots were captured from the live Replit deployment during group testing sessions.

Appendix C – Glossary

Term	Definition
Computer Vision	A field of artificial intelligence enabling computers to interpret and understand visual information from images or video.
Image Classification	A machine learning task where a model assigns a category label to an input image based on learned patterns.
TensorFlow.js	An open-source JavaScript library for training and deploying machine learning models in the browser.
MobileNet	A lightweight convolutional neural network architecture designed for efficient inference on mobile and edge devices.
Microsleep	An involuntary episode of sleep lasting from a fraction of a second to several seconds, often occurring without the driver's awareness.
False Positive	An incorrect alert generated when the driver is actually alert (i.e., the system incorrectly classifies an alert driver as fatigued).
False Negative	A missed detection where the system fails to alert when the driver is genuinely fatigued.